

Takaji Robson

[LinkedIn Profile](#)

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Research Engineer with a strong foundation in applied physics, hardware and software development, and scientific troubleshooting. Specialized in perovskite solar cells, thin-film semiconductors, and photovoltaic device physics. Experienced in designing and operating custom R&D test systems, including rapid prototyping of hardware and PCBs for renewable energy applications. Adept at developing scientific software, automating research flows, and leveraging AI tools to accelerate data analysis and advance sustainable energy technologies from concept through validation.

EDUCATION

Middlebury College, Middlebury, VT | B.A. Physics

Relevant Coursework: Experimental Physics II, Quantum Mechanics, Computational Physics, Numerical Linear Algebra, Electromagnetism, Analytical Mechanics, Waves and Fourier Analysis, Real Analysis, Statistical Mechanics

Languages: Intermediate Japanese (N3), Advanced Portuguese, Intermediate Spanish

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL, MATLAB, Bash, R

Scientific Computing and Machine Learning: NumPy, SciPy, Pandas, Matplotlib, Jupyter, Pytorch, PyVISA

Laboratory Instrumentation: Nano-Xray Diffraction, Xray Fluorescence, Phytography, SMUs, Scanning Electron Microscope

Software Development: Flask, Django, REST APIs, Docker

AI-Assisted Development & Research Tools: Claude, MCP (Model Context Protocol), LLM-assisted software development

Hardware, Engineering, & Cloud Platforms: Arduino, Raspberry Pi, ESP32, Linux, AWS, Azure, Oracle, Grafana, SolidWorks, AutoCAD, Revit

RESEARCH EXPERIENCE

Argonne National Laboratory, Lemont, IL

Research Intern (SULI) – X-Ray Science Division (Sector 19) | Summer 2026

TITLE: High-Resolution Structural Mapping of Perovskite Thin Films. Resolving (111) and (001) Microstructures via Automated Nano-XRD Analysis and a CLI-Based Computer Vision Pipeline.

- Developed the ANLCVEEVOLVE python package to automate data compression, Bragg angle alignment, peak finding, feature contouring, GUI visualization, and automated report creation for large nanoscale X-Ray Diffraction datasets.
- Implemented AI-assisted workflow automation to streamline experimental data processing, improve reproducibility, and accelerate laboratory workflow.
- Investigated structural degradation of perovskite crystals during Nano-XRD measurements by comparing the (001) and (111) crystal planes, identifying the preferred crystal orientations and feature sizes across different photovoltaic devices.

Verde Technologies, Burlington, VT

Junior R&D Engineer | April 2025 – May 2026

- Designed and deployed automated solar testing systems (32 indoor/16 outdoor) integrating custom PCBs, sensors, and data acquisition systems for multiplexed maximum power point (MPP) tracking and accelerated stress testing.
- Trained Convolutional Neural Networks (CNNs) to detect defects in photoluminescent images of perovskites thin films.
- Developed a Django-based WebUI to accelerate experimental data collection, increasing Azure database throughput Azure by 25x while reducing analysis time by over 50%.
- Deployed a cloud-based database viewer integrating an LLM-powered natural language interface to generate SQL queries, improving experimental data accessibility across the R&D team.
- Regularly presented experimental analysis, results, and industry research to the Founder, CEO, R&D Team, and Headrick Thin Film Research Group incorporating feedback to iteratively improve research tools..
- Calibrated solar simulators, configured relay systems, and integrated Keithley Source Measure Units (SMUs) using PyVISA acrossLinux and Windows environments.
- Maintained, repaired, and troubleshoot lab instrumentation, including an X-ray diffractometer and glove boxes.

Middlebury College Middlebury, VT | 2022 - 2026

Senior Thesis: An Introduction to the Scattering Method for Green's Functions in Separable Geometries

- Authored a rigorous treatment of Green's functions for analytically complex coordinate systems, deriving the Green's function for conical geometries using the eigenfunction expansion (scattering) method.
- Applied the Sommerfeld-Watson transformation and Kontorovich-Lebedev integral representations to transform discrete angular momentum sums into continuous spectra for wedge and conical geometries.
- Explored perturbation theory and hybrid semi-analytical computational methods combining analytical Green's functions with finite difference methods to model wave scattering in realistic, non-separable shapes like deformed quasi-conical tips.

Middlebury College Middlebury, VT | 2022 - 2026 (Continued)

Householder QR Decomposition Derivation and Applications Presentation – Numerical Linear Algebra | March 2024 – April 2025

- Led a MATLAB study of the Householder QR algorithm, covering derivation, worked examples, stress testing across matrices of varying size and condition number, explored relevant uses, key insights, and recent advances in a lecture-style presentation.
- Presented workflow of EKG signal processing from deconstruction, signal acquisition and transmission to reconstruction and analysis for wearable devices.

Confocal Laser Scanning Microscope – Middlebury Physics Laboratory

- Designed and constructed a prototype confocal laser scanning microscope capable of resolving 0.03 mm features using stepper motor raster scanning.
- Integrated optical components, motion control systems, and Python-based image processing with ray tracing analysis.
- Evaluated and presented imaging performance metrics, quantified error margins, and proposed improvement pathways for future 3D imaging.

PROFESSIONAL EXPERIENCE

Nuvve Holding Corp, Osaka, Japan

Software Engineering Intern | Jun 2023 – Sep 2023

- Adapted and localized an Amazon Web Services (AWS)-based staging environment from U.S. specifications to facilitate secure and scalable battery system integration into Japanese public energy markets.
- Developed and tested Java-based communication software utilizing a RabbitMQ message broker to remotely monitor and control stationary batteries and support real-time charge/discharge commands.
- Engineered a Raspberry Pi simulator to generate realistic operational data streams (voltage, current, and state of charge (SoC)), for communication protocol validation.
- Developed a Grafana-based dashboard for real-time visualization of battery parameters, communication status, and system performance.

ACTIVITIES

MiddRuns Club — Treasurer & Run Leader | Fall 2024 – May 2026

- Manage club funds and organize weekly running events (180+ members).

Additional Activities & Clubs: Middlebury Makerspace Hydroponics Team, Jiu-Jitsu Club, Chess Club, Club Basketball